Multivariate Analysis of Physiological Measurements of Elite Athletes

1 Introduction

1.1 Context and aim

This report analyses physiological measurements of female elite athletes from 5 sports using an Australian Institute of Sport dataset. The focus is on examining separation between sports using multivariate techniques including Canonical Variate Analysis (CVA), followed by applying discriminant analysis to classify two uncategorised athletes and testing prediction ability based on this data.

The need for multivariate techniques comes from the fact many variables are correlated and no single measurement clearly separates the sports.

1.2 Guide to the data (acronyms used)

This dataset contains data on 13 variables for 50 female athletes. The table below summarises all variables along with giving acronyms that will be used throughout this report.

Table 1: Description and Definitions of variables

Variable	**Full name(unit)**	**Definition**
rcc	Red cell count	Number of red blood cells in the blood.
wcc	White cell count	Number of white blood cells in the blood.
hc	Haematocrit (%)	Percentage of blood made up of red cells.
hg	Haemoglobin (g/dL)	Protein in red cells that carries oxygen.
ferr	Ferritin (µg/L)	Measure of stored iron in the body.
bmi	Body Mass Index (kg/m^2)	Ratio of weight to height squared.
ssf	Sum of skinfolds (mm)	Total thickness of seven skinfold measurements.
pcBfat	Body fat percentage (%)	Proportion of body weight that is fat.
lbm	Lean body mass (kg)	Weight of muscle and other non-fat tissue.
ht	Height (cm)	Standing height of the athlete.
wt	Weight (kg)	Total body weight.
sport	Sport category (categorical)	Athlete's sporting discipline.

Note: The five sport categories are Basketball (B_Ball), Gymnastics (Gym), Rowing (Row), Swimming (Swim), and Track–Sprinting (T_Sprnt).
Note: All athletes in the dataset are female; sex is therefore not included as a variable.

Table 2: Number of individuals per sport

Total	B_Ball	Gym	Row	Swim	T_Sprnt
50	12	3	22	9	4

2 Exploratory data analysis

Carrying out exploratory analysis provides insight into the structure of the data. It is used to inform the decisions on what methods are most appropriate moving onto later analysis,and in this case assess variables on potential grouping ability.

2.1 Summary statistics: Means and distributions

Table 3 shows that all variables except WCC and ferritin have statistically significant ANOVA p-values. This indicates that mean differences exist between sports for most measures. These differences suggest avenues for separation, while WCC and ferritin provide little evidence of between-sport variation.

The sports have high variability in body-composition variables (ssf, pcBfat, lbm, ht, wt), where Rowers and Basketball players display large values, and gymnasts show much smaller values. Track Sprinters show relatively high haematological values (rcc, hc, hg), though their small sample size limits interpretation.

The trends match up to the sporting demands expected of each, with larger stature required for Rowing's power demand and Basketballs height demand, vs smaller stature in Gymnastics, and strong Oxygen-transport markers in Sprinting. However there is substantial overlap, especially with swimming which doesn't deviate far from the sample mean, so there is no clear separating variable.

Table 3: Overall and sport-specific mean values for physiological variables

Variable	**Overall**	**B_Ball**	**Gym**	**Row**	**Swim**	**T_Sprnt**	**ANOVA p-val**
rcc	4.42	4.34	4.23	4.46	4.29	4.93	$5.68e^{-04}$
wcc	6.61	6.18	6.77	6.76	6.21	7.80	$3.19e^{-01}$
hc	40.97	39.64	39.27	41.59	40.43	44.03	$5.14e^{-03}$
hg	13.69	13.05	13.47	14.03	13.57	14.18	$8.80e^{-03}$
ferr	52.04	46.25	66.00	50.41	63.44	42.25	$3.57e^{-01}$
bmi	21.85	21.36	18.76	22.75	21.89	20.59	$1.32e^{-03}$
ssf	82.85	97.63	48.00	93.20	65.62	46.48	$7.49e^{-06}$
pcBfat	17.54	20.01	10.91	19.79	13.89	10.90	$3.69e^{-08}$
lbm	56.22	57.35	38.78	58.42	56.50	53.21	$8.22e^{-06}$
ht	176.65	183.37	152.27	178.86	173.18	170.48	$2.22e^{-08}$
wt	68.59	72.03	43.57	72.90	65.73	59.73	$4.68e^{-07}$

Note: Sport categories are Basketball (B_Ball), Gymnastics (Gym), Rowing (Row), Swimming (Swim), and Track–Sprinting (T_Sprnt).

Most variables are approximately normally distributed, with some mild skewness in ferritin (and others) or slightly bimodal with BMI (Figure 1). These deviations do not hinder the use of multivariate methods but indicate that univariate separation will be limited.

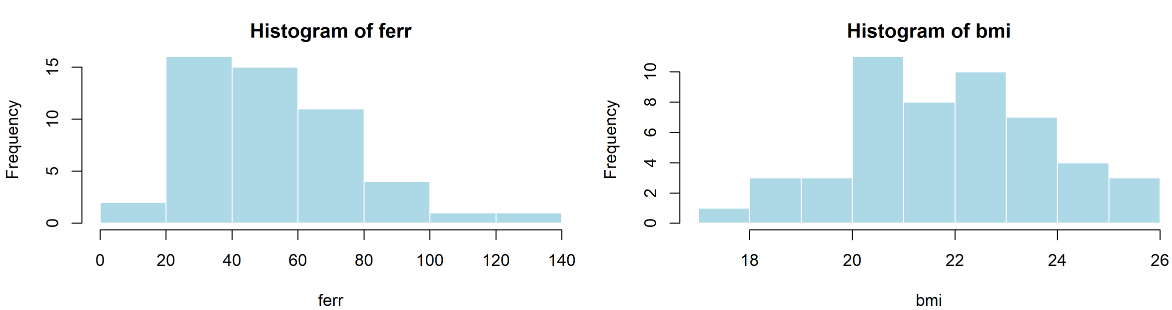

Figure 1: Histograms: Examples that differ from the norm

2.2 Segment glyphs

The trends seen in Table 3 can be seen at a glance in Figure 2. The red colour scheme indicates the haematological values and blue colour scheme indicates body-composition variables. In a qualitative way this provides the same information for increased familiarity with the data.

The larger the segment the higher the value. These have been min-max normalised then a constant added so that relatively small values are still visible.

The correlation within body-composition and haematological variables can be seen where the blue segments or red segments are usually all of a similar scale in each individual. For example the size comparison between gymnasts and every other sport.

These are the aspects seen in pervious and future tests, however what is unique to this plot is seeing the individuals variety. Outlier individuals can be identified, for example swimmers 42 and 43 are comparatively small in the group. There are also visible individuals with particularly small haematological values (B_Ball 1,3 & 4).

This can be a good place to refer back to when context is required for interpretation.

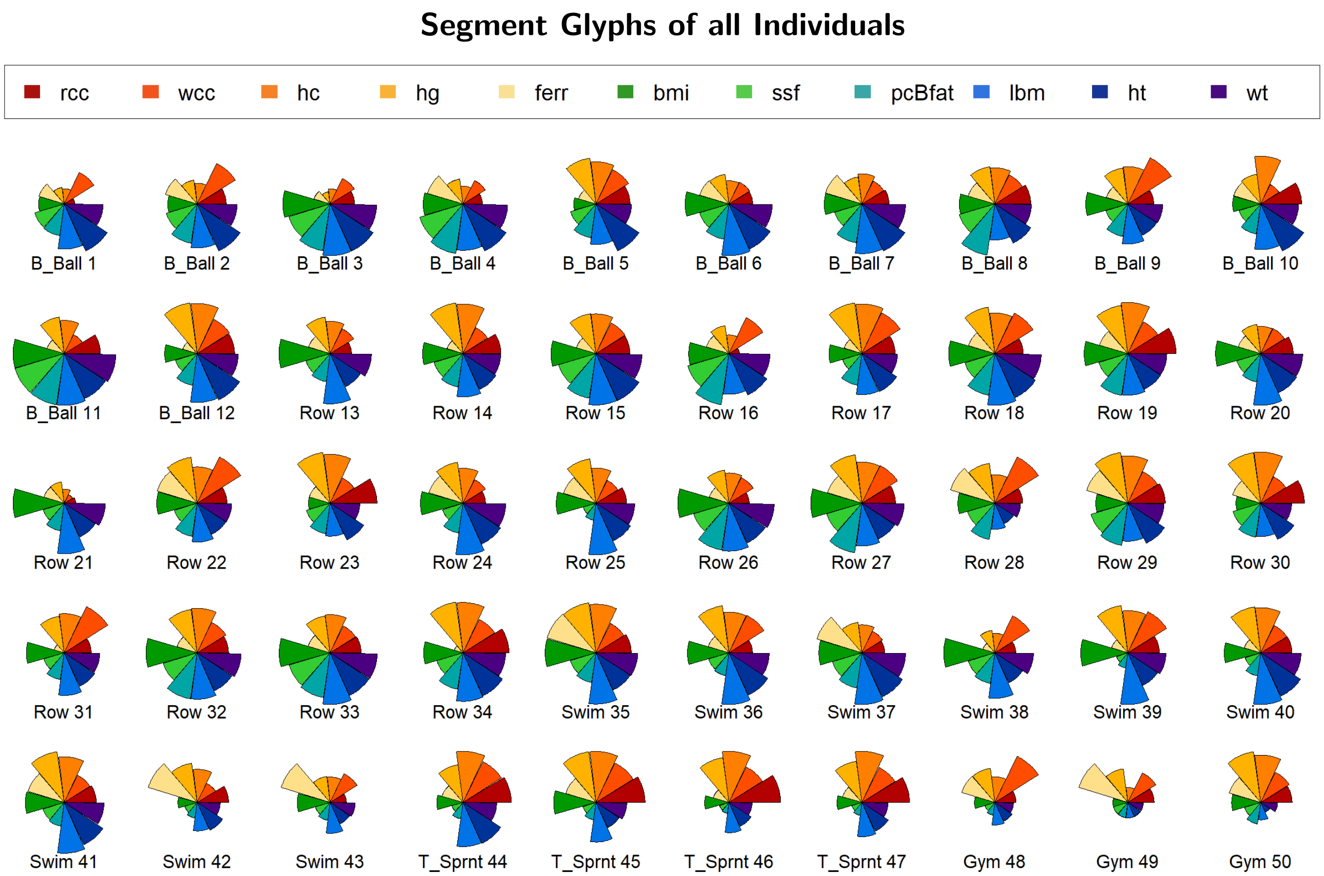

Figure 2: Segment glyphs: For a visual representation of the individuals characteristics

2.3 Pairs plots

This is a plot of each variable plotted against every other variable in a set of pairs. The points are coloured by sport. This is used to visualise bivariate interaction across the whole data set at once.

Bivariate relationships (Figure 3) highlight strong collinearity within haematological variables and body-composition separately. These Bivariate correlation suggests an underlying physiological factor is being explained by multiple variables. These redundant dimensions can be combined in techniques such as principal component analysis and canonical variate analysis.

Despite some group outliers (swim in the ferr biplots, and purple in lbm biplots) the majority of individuals overlap regardless of sport. Once again, as there is no clear separating variable, the need for multivariate methods of dimension reduction are clear.

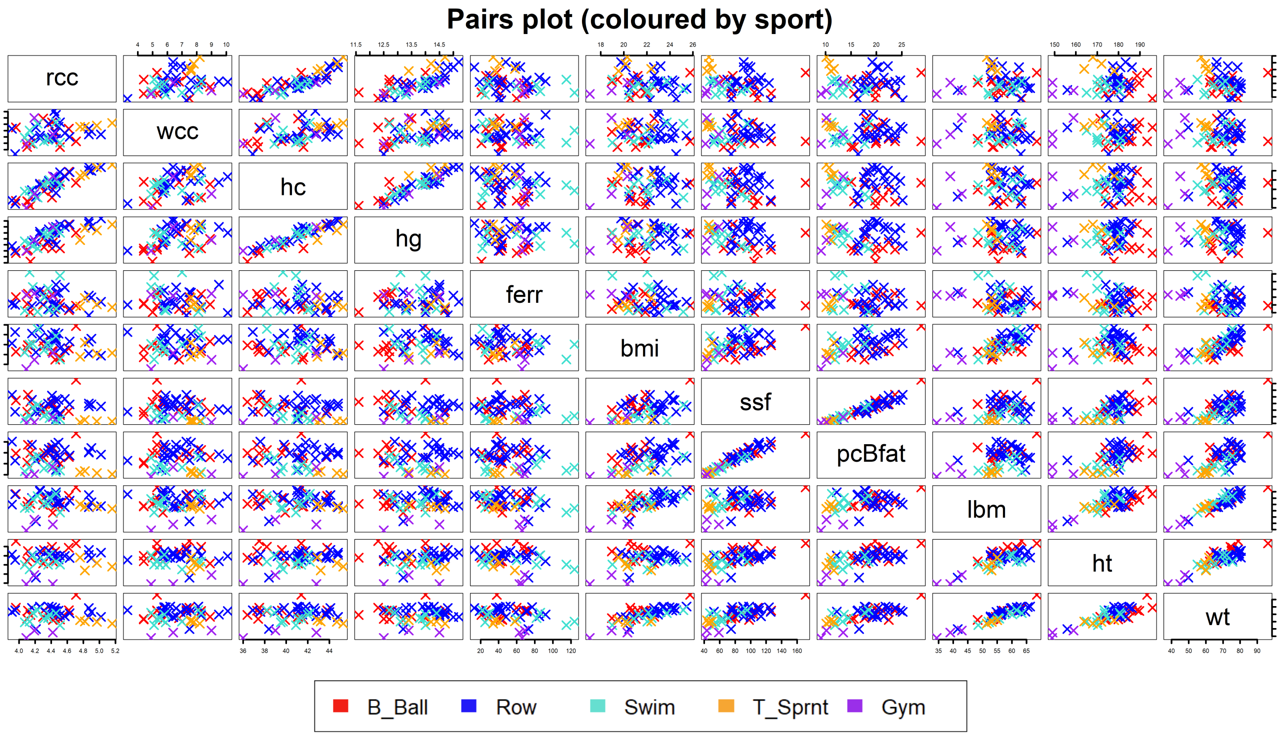

Figure 3: Bivariate plots showing both general distribution and correlations

3 Examining separation

3.1 Chosen method and explanation

Canonical Variate Analysis (CVA) constructs linear combinations of the original variables that maximise the ratio of between-group to within-group variance. The output is axes that optimally separate the predefined groups. PCA maximises overall variance without considering the predefined groups. Given the substantial collinearity in the data, limited visible separation and the aim of separation into existing groups, CVA is the most appropriate method in this instance.

3.2 In practice and observations

Carrying out the CVA produces 4 Canonical variates. The first 2 are chosen as our new axes as they account for 81.76% of the cumulative variation. The third CV could be included but it only accounts for a very small amount of variance. For the sake of visualisation 2 is sufficient.

Table 4: Canonical variate coefficients

	rcc	**wcc**	**hc**	**hg**	**ferr**	**bmi**	**ssf**	**pcBfat**	**lbm**	**ht**	**wt**
CV1	0.3925	-0.2105	-0.3283	0.7415	-0.0007	-0.0771	-0.0121	1.4595	1.6853	0.0415	-1.3327
CV2	-3.6678	-0.0452	-0.1096	1.3031	-0.0052	-6.0109	-0.0065	1.5069	1.8620	-1.4432	0.3352

Interpretation is based on the relative size and sign of the CV coefficients:

CV1 accounts individually for 54.8% of the variance, and the variable is most strongly influenced by the body-composition factors (pcBfat, lbm, wt) as the coefficients are large positive numbers.

CV2 accounts for 26.9% of variance, and is strongly negatively influenced by more body-composition factors, BMI with a coefficient of -6.0 and positively by lbm to a lesser extent.

There are negligible contributions to CV1 and CV2 by the other variables, this indicates that most of the between group separation is explained by body-composition factors.

CV	Eigenvalue	Cumulative%
CV1	4.41	54.82
CV2	2.17	81.76
CV3	0.97	93.68
CV4	0.51	100

Table 5: Eigenvalues and explained variance

3.3 CVA biplot

This plot visualises the separation (or lack of separation) achieved by the CVA process. An X marks the centroid of each sport on the plot, and the colour coded ellipse represents 1 standard deviation away from the centroid. The arrows denote the direction and scaled magnitude of each variables influence on the scores of the canonical variables.

The small Gymnastics group has separated extremely well, although with a large s.d. Track Sprint is the only other group to be fully separated and with a relatively small s.d.

Around the origin of the plot we see the Rowers and the Basketball players overlap so much that the centroids are within 1 s.d. in this plot, so unfortunately they have not been successfully separated. The Swim group does also overlap with Rowing and Basketball significantly but not within 1 s.d.

Once again these groups are separating along similar lines as in the exploratory analysis. Though there was enough separation that it is worth attempting to classify uncategorised new data.

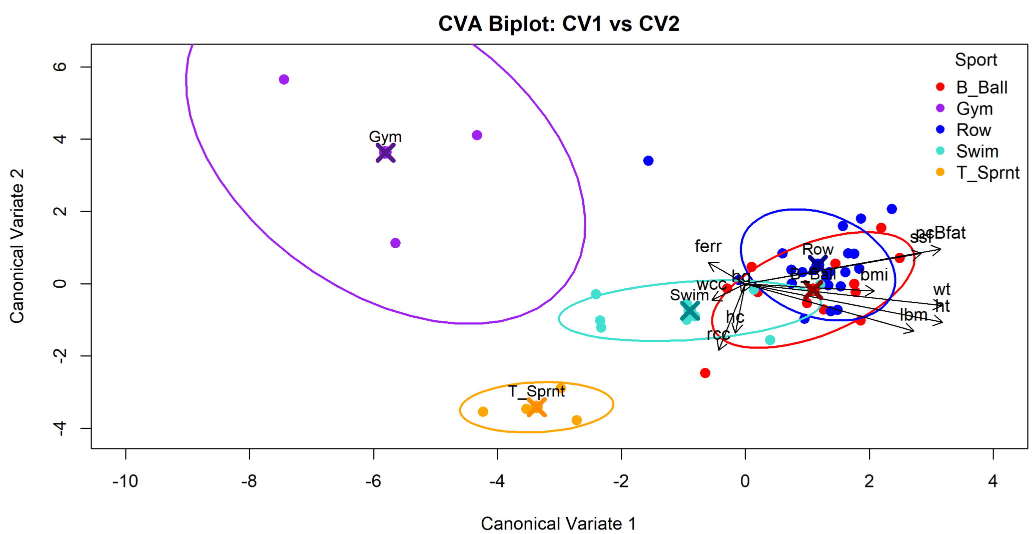

Figure 4: Biplot of CV1 and CV2

4 Categorising new individuals

Here is the data for 2 new athletes. This contains all the same variables as the original data set, except the sport variable.

	rcc	wcc	hc	hg	ferr	bmi	ssf	pcBfat	lbm	ht	wt
Athlete 1	4.35	7.8	41.4	14.1	30	22.03	117.8	23.3	48.32	169.1	63
Athlete 2	4.53	5.00	40.7	14.0	41	17.79	56.8	12.55	38.3	156.9	44.8

Table 6: Physiological measurements for the two new athletes.

In this section, Linear Discriminant analysis, otherwise known as Canonical variate analysis, will be used to categorise each athlete into a sport group.

The canonical variates from CVA coincide with the discriminant functions for these groups, so the CV1 and CV2 scores are the required discriminant scores going forward.

The scores of the discriminant function are obtained by calculating:

$$y_k = \sum_{j=1}^{p} a_{kj}(x_j - \bar{x}_j),$$

where a_{kj} are the raw canonical coefficients and $(x_j - \bar{x}_j)$ are the mean–centred measurements.

4.1 Score hand calculation

1. Mean–centre the athlete's data. Athlete 2 has

$$\mathbf{x}^* = (4.53,\ 5.00,\ 40.7,\ 14.0,\ 41,\ 17.79,\ 56.8,\ 12.55,\ 38.3,\ 156.9,\ 44.8).$$

The exact variable means are

$$\bar{\mathbf{x}} = (4.4218,\ 6.6060,\ 40.9680,\ 13.6900,\ 52.0400,\ 21.8508,\ 82.8480,\ 17.5372,\ 56.2230,\ 176.6520,\ 68.5860).$$

Thus the centred vector is

$$\mathbf{z}^* = \mathbf{x}^* - \bar{\mathbf{x}} = (0.1082,\ -1.6060,\ -0.2680,\ 0.3100,\ -11.0400,\ -4.0608,\ -26.0480,\ -4.9872,\ -17.9230,\ -19.7520,\ -23.7860).$$

2. Compute CV1 score. The raw CV1 coefficients are

$$\mathbf{a}_1 = (0.3925,\ -0.2105,\ -0.3283,\ 0.7415,\ -0.0007,\ -0.0771,\ -0.0121,\ 1.4595,\ 1.6853,\ 0.0415,\ -1.3327).$$

Hence

$$\begin{aligned} y_1^* &= 0.3925(0.1082) + (-0.2105)(-1.6060) + (-0.3283)(-0.2680) + 0.7415(0.3100) \\ &\quad + (-0.0007)(-11.0400) + (-0.0771)(-4.0608) + (-0.0121)(-26.0480) \\ &\quad + 1.4595(-4.9872) + 1.6853(-17.9230) + 0.0415(-19.7520) + (-1.3327)(-23.7860) \\ &= 0.0425 + 0.3388 + 0.0880 + 0.2299 + 0.0077 + 0.3131 + 0.3158 - 7.2781 - 30.1887 - 0.8207 + 31.7060 \\ &= \boxed{-5.2467}. \end{aligned}$$

3. Compute CV2 score. The raw CV2 coefficients are

$$\mathbf{a}_2 = (-3.6678,\ -0.0452,\ -0.1096,\ 1.3031,\ -0.0052,\ -6.0109,\ -0.0065,\ 1.5069,\ 1.8620,\ -1.4432,\ 0.3352).$$

Hence

$$\begin{aligned}
y_2^* &= (-3.6678)(0.1082) + (-0.0452)(-1.6060) + (-0.1096)(-0.2680) + 1.3031(0.3100) \\
&\quad + (-0.0052)(-11.0400) + (-6.0109)(-4.0608) + (-0.0065)(-26.0480) \\
&\quad + 1.5069(-4.9872) + 1.8620(-17.9230) + (-1.4432)(-19.7520) + 0.3352(-23.7860) \\
&= -0.3970 + 0.0726 + 0.0294 + 0.4039 + 0.0574 + 24.4126 + 0.1693 - 7.5199 - 33.3867 + 28.5071 - 7.9687 \\
&= \boxed{4.3801}.
\end{aligned}$$

4. Final canonical coordinates. We have our final canonical scores for Athlete 2 :

$$(\text{CV1, CV2}) = (\boxed{-5.2467}, \boxed{4.3801}),$$

which match the values returned by the R code to 2s.f. This is obviously closest to the Gymnastics centroid (Figure 5), and very far away from the others so can be classified fairly confidently in the Gymnastics group.

4.2 New individuals scores plotted

Athlete 1. has canonical scores $(\text{CV1}, \text{CV2}) = (1.68,\ 2.50)$, plotted in pink as New1. This is closest to the Rowing centroid despite not being within 1 s.d, so could be classified as such. However due to the major overlaps of Rowing, Basketball and Swimming this athlete cannot be confidently placed.

Athlete 2. has canonical scores $(\text{CV1}, \text{CV2}) = (-5.25,\ 4.38)$, plotted in green as New2. This athlete is within 1 s.d of the Gymnastics centroid and very far from the other 4 groups, so can be confidently grouped with the gymnasts. This could still not be correct as this athlete may not even compete in the 5 sports within this report, so could be entirely incorrectly classified.

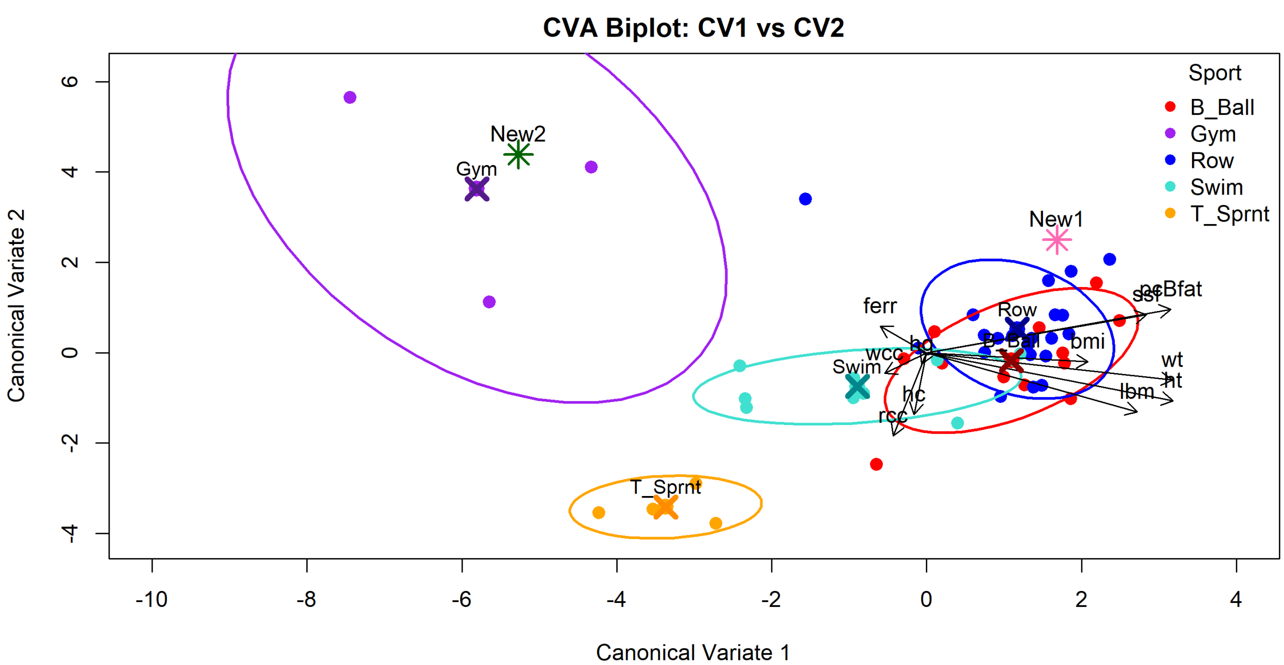

Figure 5: Canonical variate scores plot with the 2 uncategorised athletes

5 Reliability tests

5.1 Obtaining APER

Actual	B_Ball	Gym	Row	Swim	T_Sprnt
B_Ball	3	0	0	0	0
Gym	0	0	0	0	0
Row	1	0	5	0	0
Swim	0	0	2	1	0
T_Sprnt	1	0	0	0	2

Table 7: Confusion matrix from 70/30 train–test LDA classification.

The apparent error rate (APER) was

$$\text{APER} = 0.267,$$

indicating that 26.7% of test-set athletes were misclassified.

5.2 Interpretation

Although the APER of 0.267 indicates moderate ability at prediction, it is unlikely to reflect the real error rate. The data is imbalanced, with very small groups such as gymnastics ($n = 3$) and Track Sprint ($n = 4$), meaning that with a 70/30 split the training set may contain only one or two of these sports. The resulting discriminant rule could be overly sensitive, and different random partitions would likely produce different APER values.

The confusion matrix also shows the overlap observed in the CVA plot, particularly between Swimming and Rowing, which limits classification reliability. Cross-validation would be a more reliable estimate for this data set, as it maximises the chance that every sport is represented appropriately in this small data set.

6 Conclusion

Strong physiological trends emerged across the analysis, yet substantial overlap persisted, so no single variable provided clear separation between sports. CVA amplified the patterns identified in the exploratory work and showed some potential for predicting group membership. However, the unreliability of the classification is largely driven by the small, imbalanced dataset and, in practice, the fact that not all sporting categories or relevant explanatory variables are represented. Realistic error rates are therefore crucial for interpreting these results. In this context, factor analysis could help identify latent variables that may be driving the unexplained variation and contributing to the remaining overlap.